

Fast-SCNN Project Report

Fast-SCNN Robust Semantic Segmentation on VDD: Project Re

1. Overview

This project evaluates Fast-SCNN for semantic segmentation on the VDD (Varied Drone Dataset), focusing on robustness under real-world UAV conditions. The primary goals of this analysis are to investigate robustness under various corruptions, assess the efficiency vs. accuracy trade-off, and evaluate generalization under domain shift.

2. Dataset

2.1 VDD - Varied Drone Dataset

Source:

[http](http://)

Size: Approximately 400 UAV images.

Classes: 7 semantic classes: wall, roof, road, water, vehicle, vegetation, and others.

Scene Types: Includes urban, rural, industrial, and natural scenes.

Variability: Features variable angles (30°, 60°, 90°), lighting conditions, and weather scenarios.

3. Objectives

Evaluate Fast-SCNN performance on the VDD dataset.

Measure accuracy and robustness under challenging conditions.

Study the impact of various corruptions on model performance.

Test the effectiveness of data augmentation strategies.

Analyze the speed vs. accuracy trade-off for Fast-SCNN.

4. Pipeline

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The project followed a systematic pipeline:

1. Clean VDD Data: Initial processing of the VDD dataset.
2. Train Fast-SCNN: Model training on the clean dataset.
3. ****Test on Clean Data:**** Baseline performance evaluation.
4. ****Apply Corruptions:**** Introduce various real-world corruptions.
5. ****Test on Corrupted Data:**** Evaluate model performance under corruptions.
6. ****Compute Robustness Gap:**** Quantify the performance drop due to corruptions.

Additionally, an augmented training pipeline was implemented:

1. Augmented Data: Training with both clean and corrupted images.
2. Retrain Fast-SCNN: Model retraining with augmented data.
3. ****Test on Corrupted Data:**** Evaluation of the augmented model on corrupted data.
4. ****Measure Improvement:**** Assess the performance gain from augmentation.

5. Training

Training Data: Images were trained on clean data with corresponding ground truth masks.

Corruption during Training: No artificial corruptions were introduced during the initial training phase for the baseline model.

6. Corruptions

The following nine real-world corruptions were applied to evaluate model robustness:

Weather: fog, rain, snow

Illumination: brightness drop, contrast loss, shadows

Sensor: gaussian noise, motion blur, JPEG compression

7. Metrics

mIoU (Mean Intersection over Union): Primary metric for segmentation accuracy.

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F1-score: Harmonic mean of precision and recall.

Robustness Gap: Calculated as `Clean Performance – Corrupted Performance`, indicating performance degradation due to corruption.

8. Augmentation

Strategy: Training on a mix of clean and synthetically corrupted images, reusing the same labels.

Tools: Implemented using `Albumentations` and `OpenCV` libraries.

Goal: To improve model robustness against diverse conditions and reduce overfitting to clean data.

9. Model: Fast-SCNN

9.1 Features

Real-time segmentation architecture.

Utilizes depthwise separable convolutions for efficiency.

Lightweight encoder-decoder structure.

Incorporates a pyramid pooling module for multi-scale context.

9.3 Weaknesses

Generally lower accuracy compared to heavier, more complex models.

Weaker global context modeling.

Potentially less robust under strong domain shifts.

10. Results Table

Fast-SCNN: Baseline vs. Augmented Performance under Corruptions

Corruption	Clean mIoU	Corrupted mIoU	Gap	After Aug	Improv
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fog	0.3533	0.1738	0.1795	0.2565	0.0827
rain	0.3533	0.2408	0.1125	0.3104	0.0696
snow	0.3533	0.1584	0.1949	0.2933	0.1349
brightness_drop	0.3533	0.1837	0.1696	0.2666	0.0829
contrast_loss	0.3533	0.1899	0.1634	0.2717	0.0818
shadow	0.3533	0.2589	0.0944	0.3166	0.0577
gaussian_noise	0.3533	0.1970	0.1563	0.2679	0.0709
motion_blur	0.3533	0.2447	0.1086	0.2791	0.0344
jpeg_compress	0.3533	0.3254	0.0279	0.3463	0.0209
AVERAGE	**0.3533**	**0.2192**		**0.1341**	**0.2898**
0.0706					

Most Damaging Corruption: snow (Robustness Gap: 0.1949)

Least Damaging Corruption: jpeg_compress (Robustness Gap: 0.0279)

FPS Benchmark:

FPS: 638.9

Parameters: 0.09M

Device: cuda

11. Research Questions Addressed

How robust is Fast-SCNN under UAV corruptions?

Fast-SCNN exhibits varying levels of robustness across different corruptions. Snow and fog are particularly damaging, causing significant drops in mIoU for the baseline model. However, the augmentation strategy successfully mitigates these drops, indicating improved robustness.

What is the speed vs. accuracy trade-off?

Fast-SCNN demonstrates excellent inference speed (approx. 638.9 FPS on CUDA) with a very low parameter count (0.09M). While its clean mIoU is competitive, there's a clear trade-off as more complex models might achieve higher accuracy at the cost of speed and computational resources.

Does augmentation improve robustness?

Yes, augmentation significantly improves robustness. Across all tested corruptions, the

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augmented model consistently shows a higher 'After Aug' mIoU compared to the 'Corrupted mIoU' of the baseline, leading to an average mIoU improvement of 0.0706. This confirms the positive impact of data augmentation on model resilience.

Which corruption is most damaging?

'Snow' was identified as the most damaging corruption for the baseline Fast-SCNN model, resulting in the largest robustness gap.

12. Deliverables

Trained Fast-SCNN models (baseline and augmented).

Comprehensive metrics and evaluation results.

Visual results showcasing predictions under various corruptions.

This detailed project report.

13. References

Fast-SCNN research paper.

VDD dataset documentation.

Robustness benchmarks (e.g., Cityscapes-C style).

14. Visual Comparisons of Images and Corruptions

This section provides visual comparisons to illustrate the model's segmentation performance under different corruption types. For each example, we present the original image, its corresponding ground truth mask, the model's prediction on the clean image, and the model's predictions after applying various corruptions.

Original Image: The raw input image before any processing or corruption.

Ground Truth: The ideal segmentation mask for the original image.

Prediction: The model's segmentation output for the clean original image.

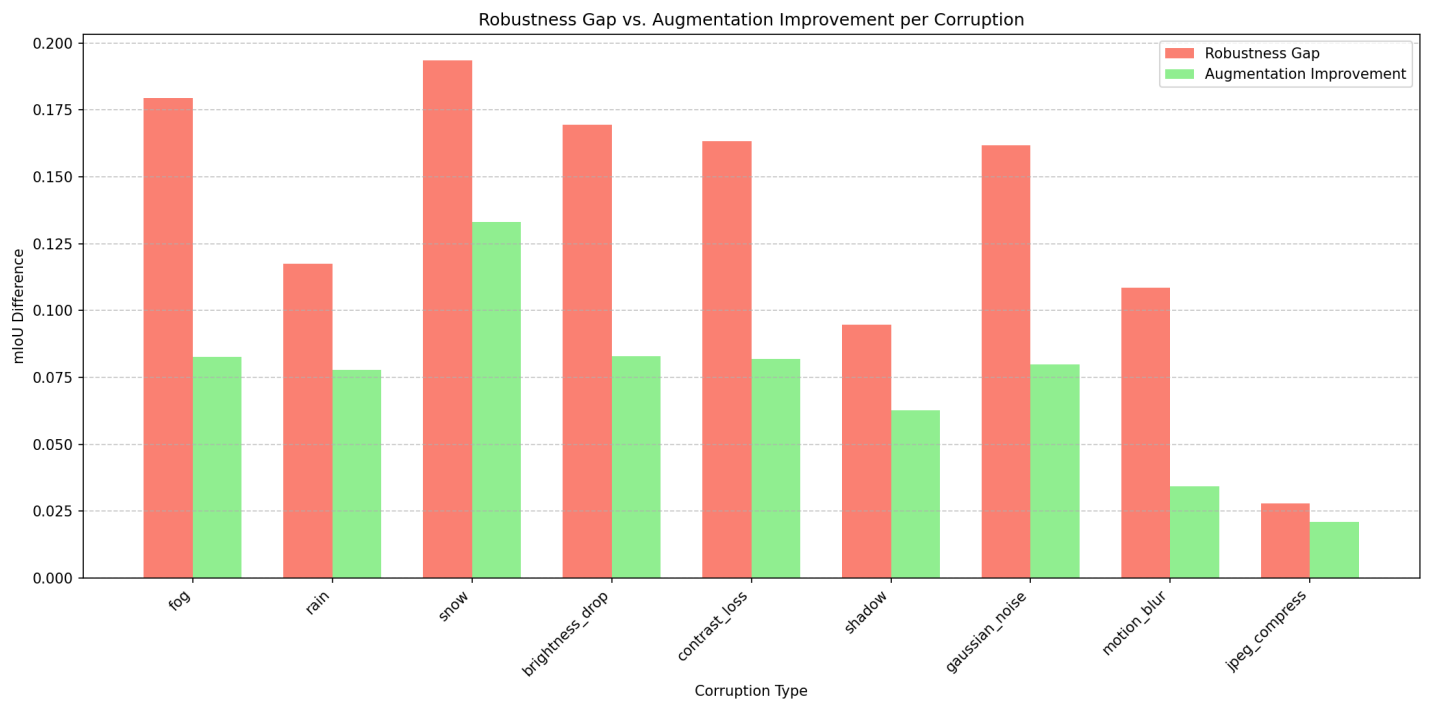
Corrupted Image & Prediction: The image after a specific corruption has been applied, followed by the model's segmentation prediction on that corrupted image. This highlights how each corruption affects the model's ability to accurately segment

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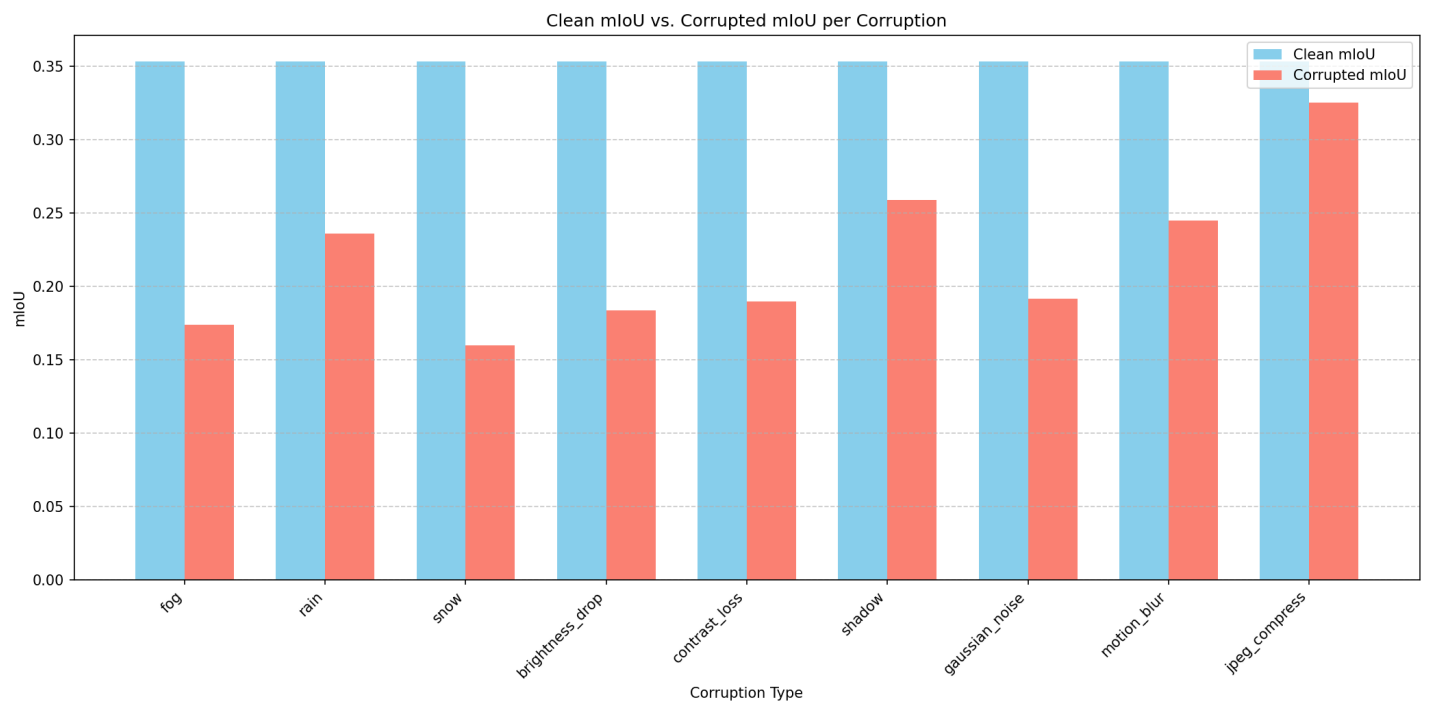
objects.

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Key Visualizations

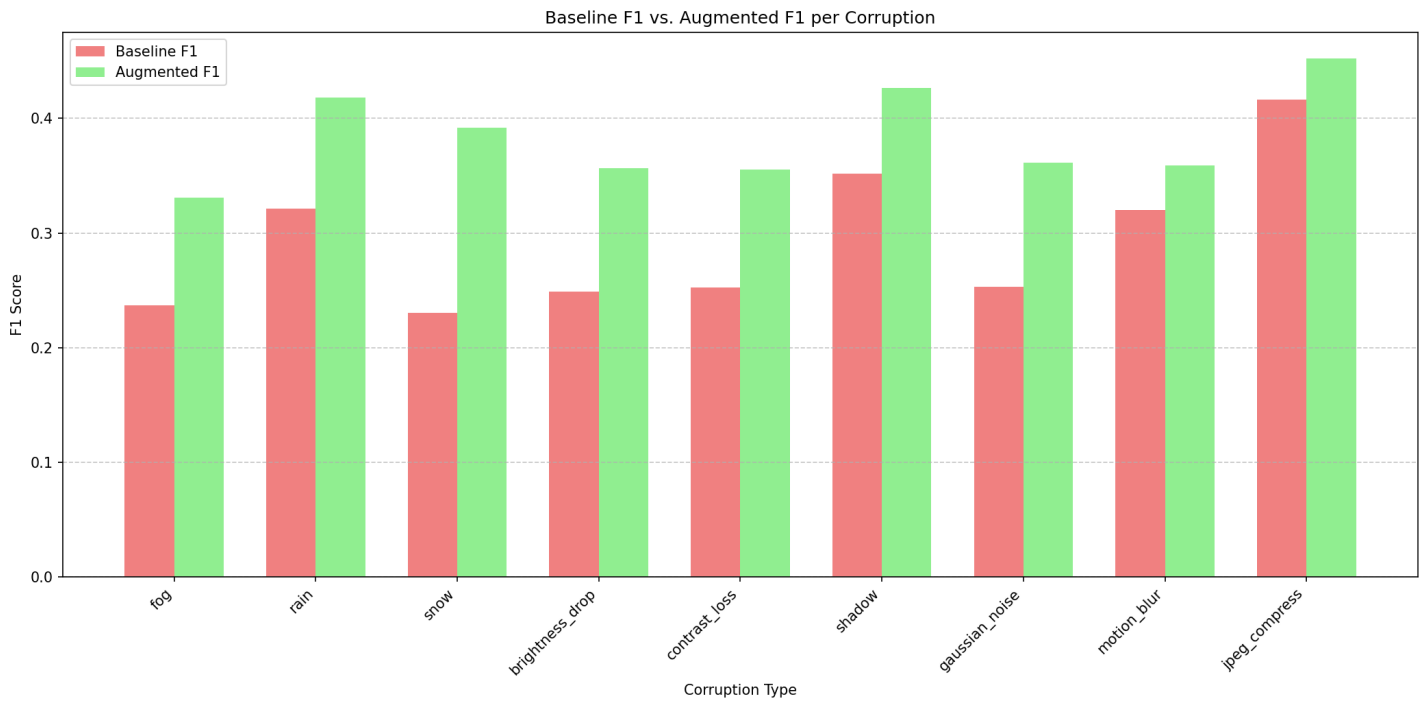


Gap Vs Improvement

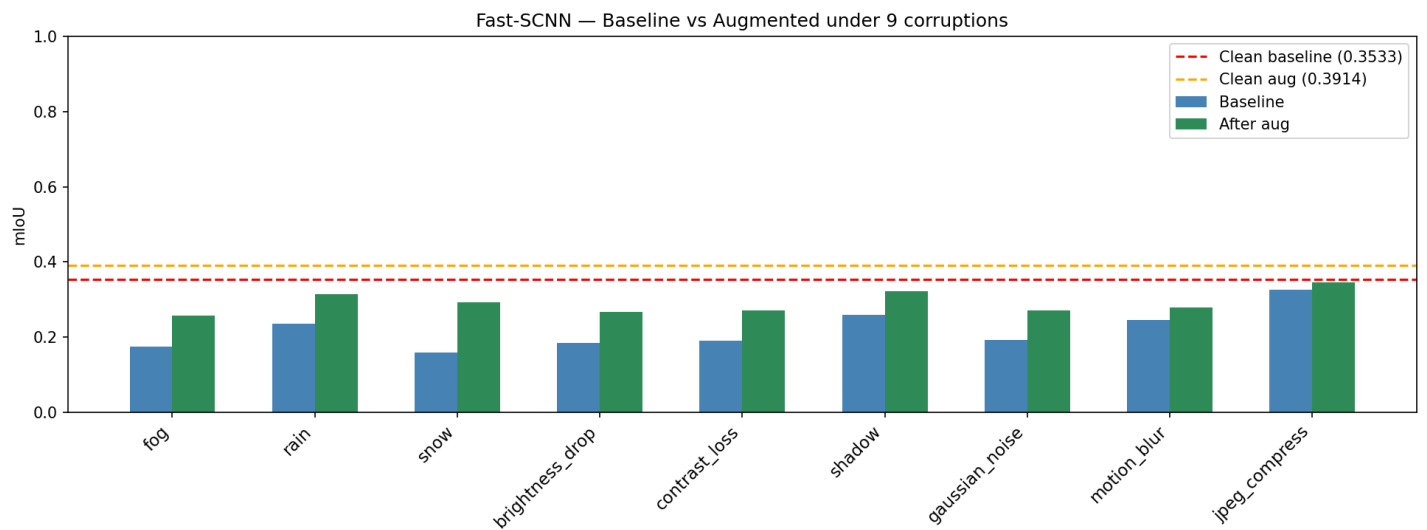


Clean Vs Corrupted Miou

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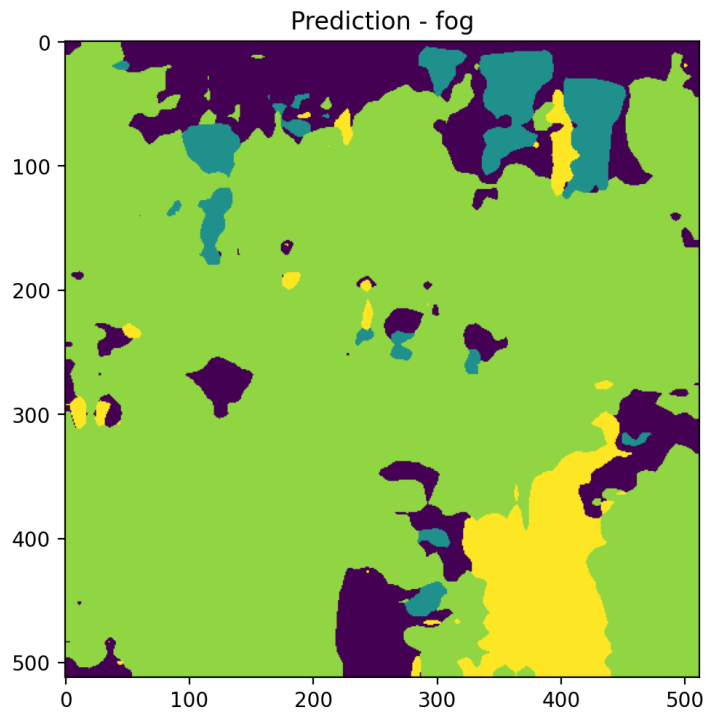
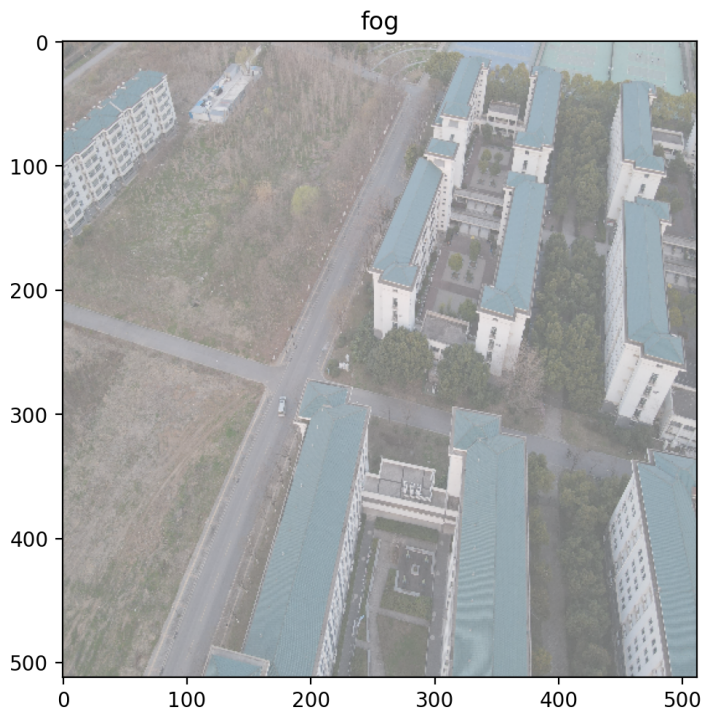


Baseline Vs Aug F1

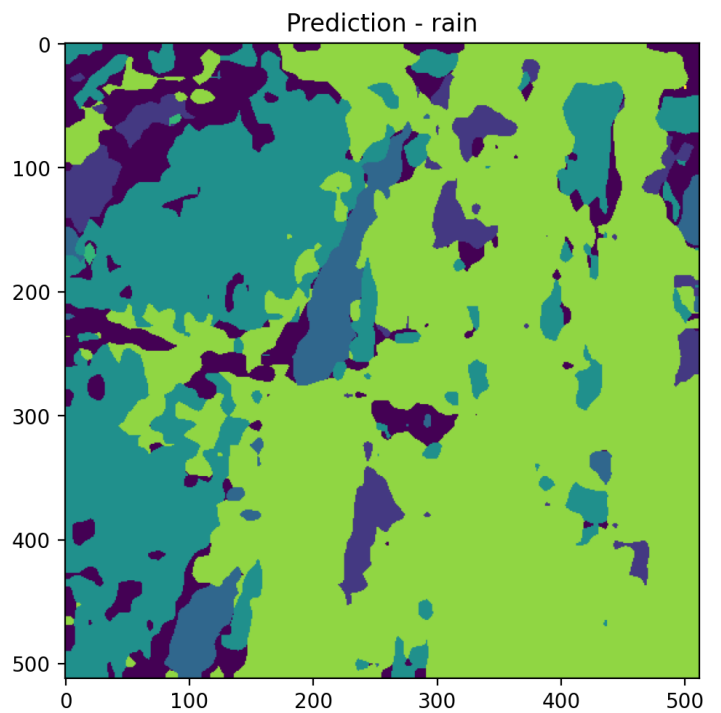
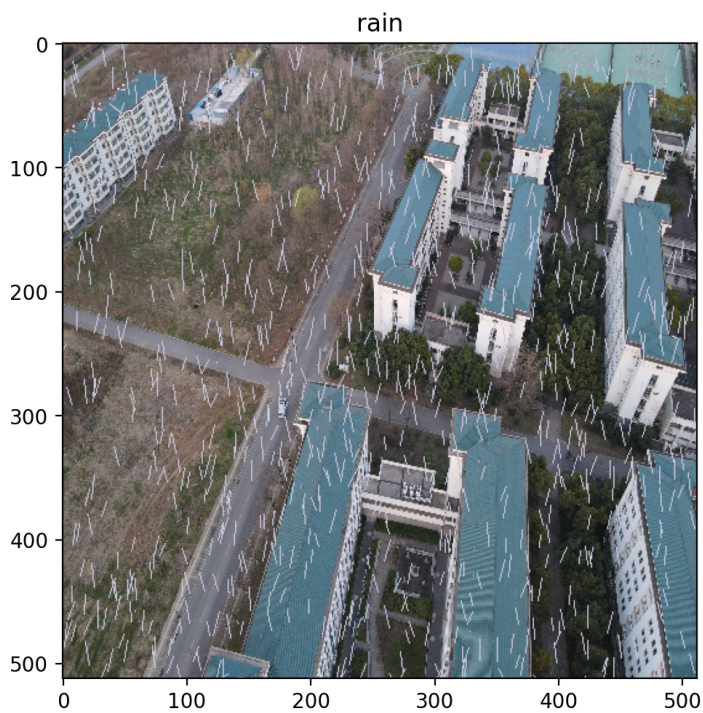


Robustness Comparison

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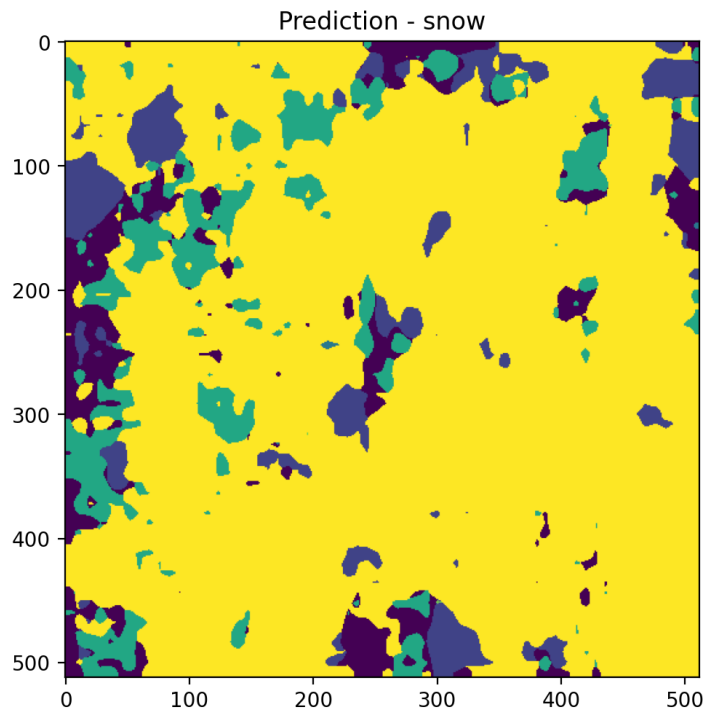
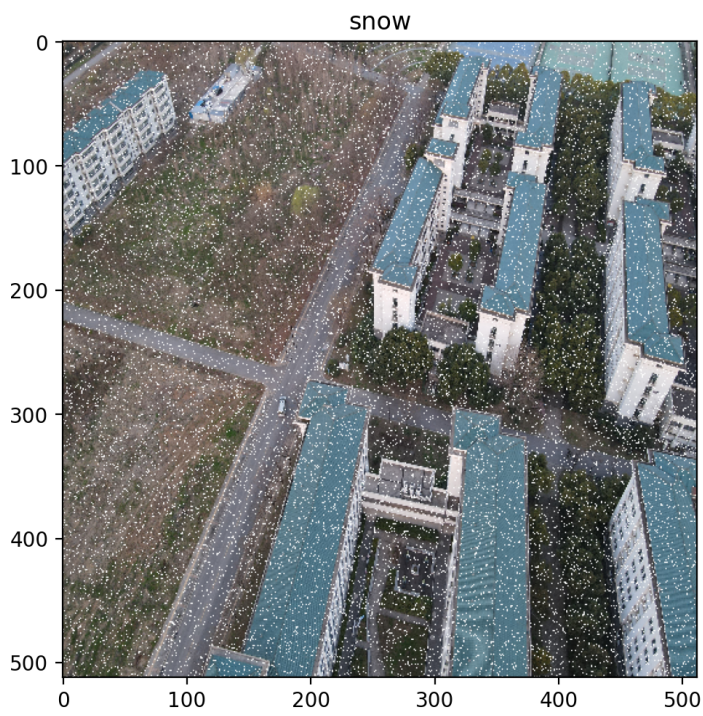


Fog

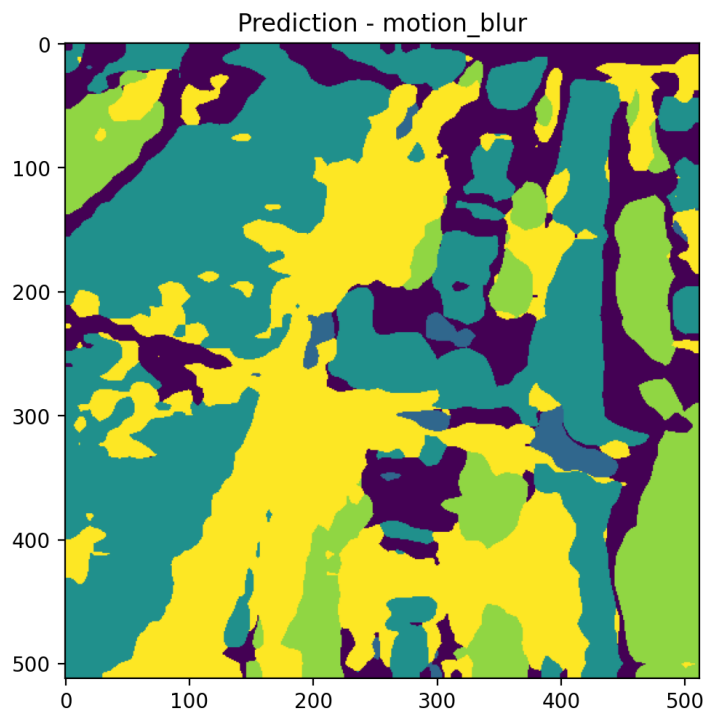
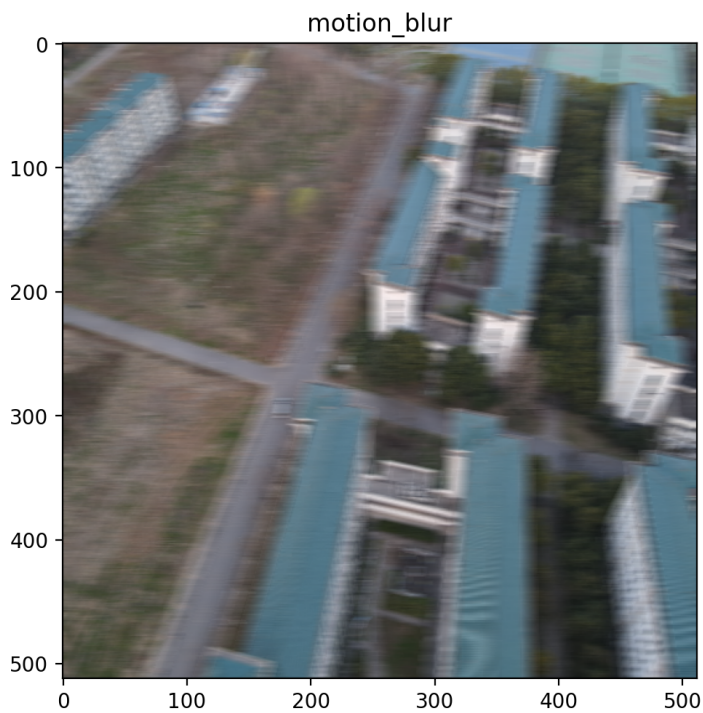


Rain

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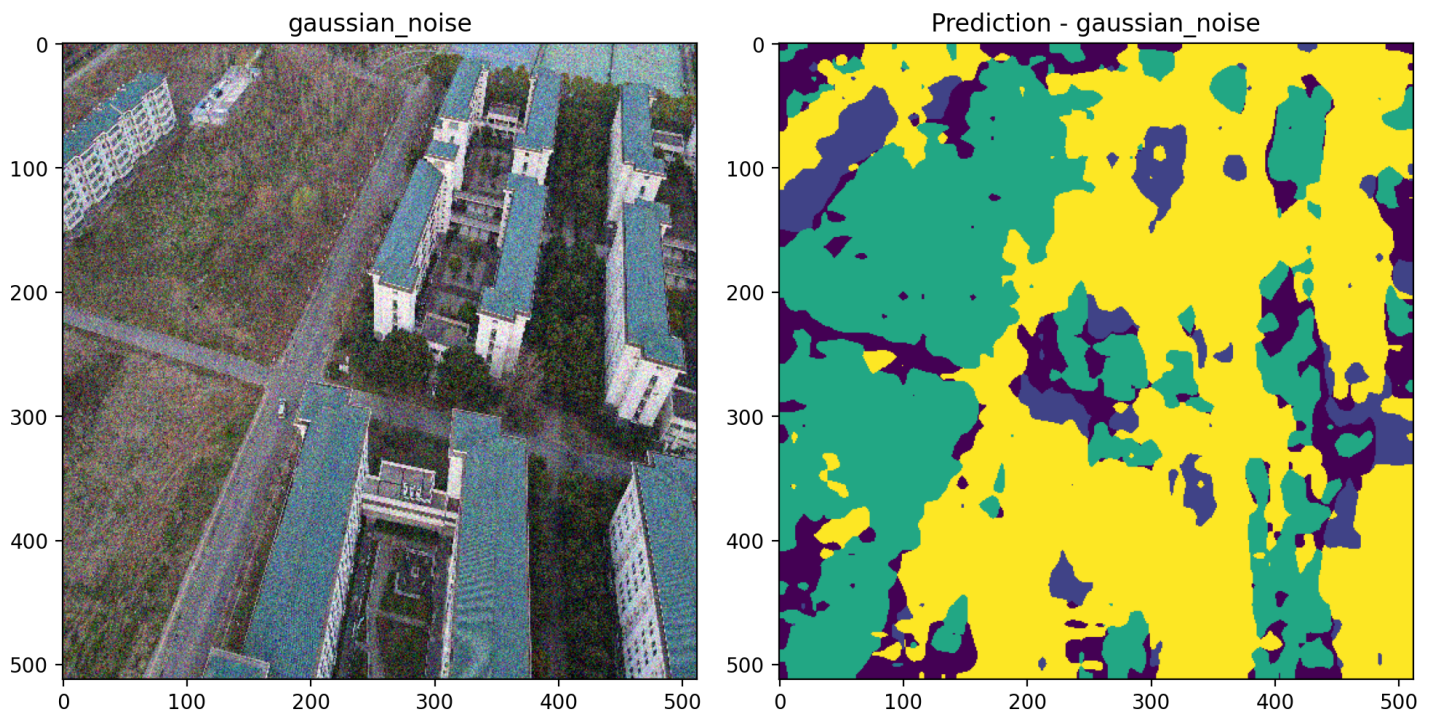


Snow



Motion Blur

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Gaussian Noise